

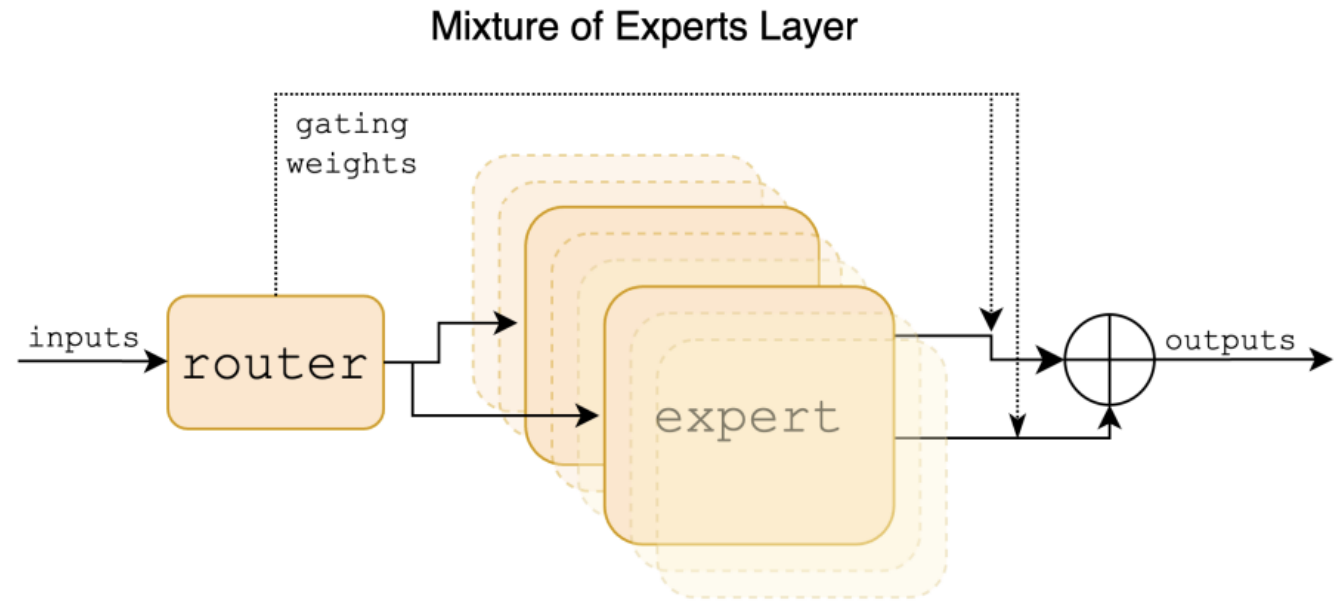


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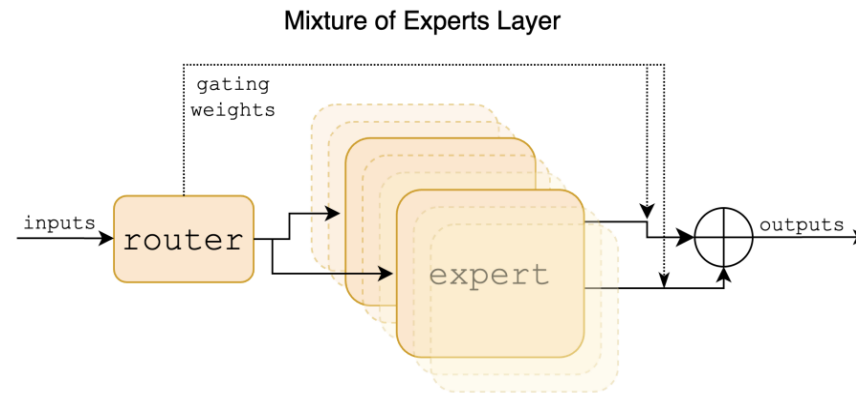
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Model Speed: Mixtral (decoder-only)

- Instead of training one large models
- A sparse mixture of experts (SMoE)
- At each layer, a router chooses 2 experts out of 8 total
- Output of the 2 experts combined additively
- Why is this efficient? Large number of experts (params) can be had but only 2 experts are run each time, improving speed despite increased params



Router



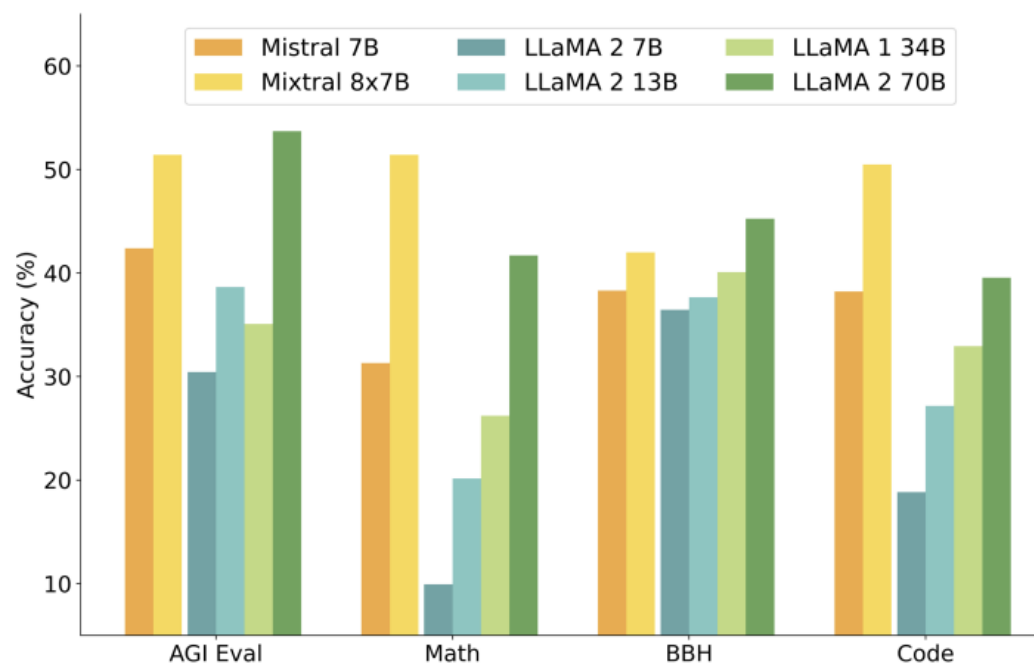
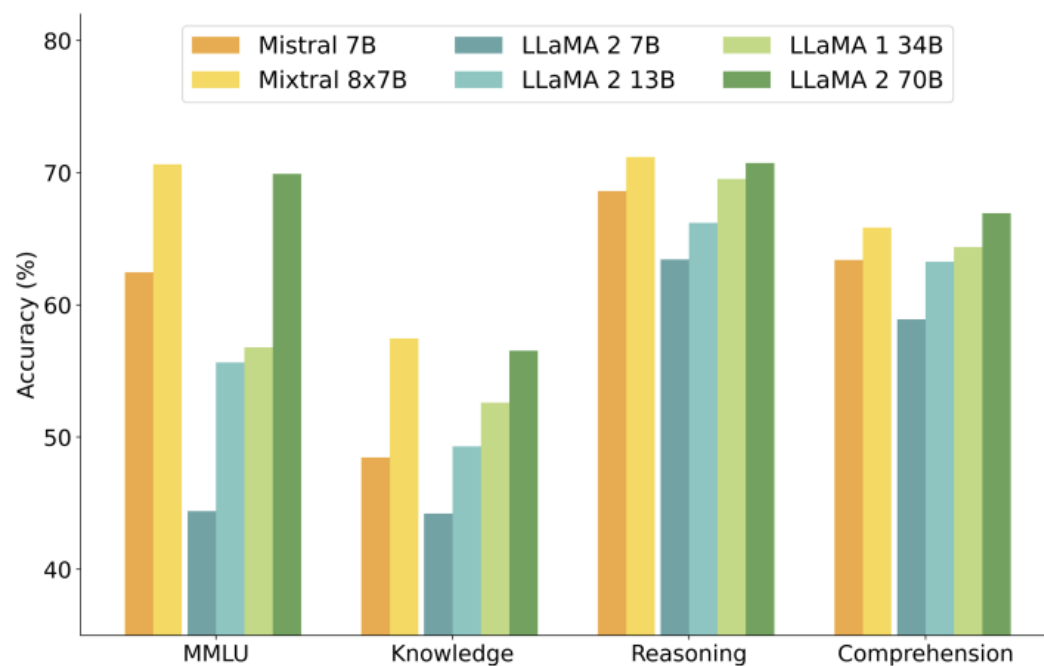
$$\{E_0, E_i, \dots, E_{n-1}\}$$

$$\sum_{i=0}^{n-1} G(x)_i \cdot E_i(x), \quad G(x) := \text{Softmax}(\text{TopK}(x \cdot W_g))$$

$(\text{TopK}(\ell))_i := \ell_i$ if ℓ_i is among the top-K coordinates of logits $\ell \in \mathbb{R}^n$ and $(\text{TopK}(\ell))_i := -\infty$

$$y = \sum_{i=0}^{n-1} \text{Softmax}(\text{Top2}(x \cdot W_g))_i \cdot \text{SwiGLU}_i(x).$$

Results



Results

Model	Active Params	MMLU	HellaS	WinoG	PIQA	Arc-e	Arc-c	NQ	TriQA	HumanE	MBPP	Math	GSM8K
LLaMA 2 7B	7B	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	17.5%	56.6%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	13B	55.6%	80.7%	72.9%	80.8%	75.2%	48.8%	16.7%	64.0%	18.9%	35.4%	6.0%	34.3%
LLaMA 1 33B	33B	56.8%	83.7%	76.2%	82.2%	79.6%	54.4%	24.1%	68.5%	25.0%	40.9%	8.4%	44.1%
LLaMA 2 70B	70B	69.9%	85.4%	80.4%	82.6%	79.9%	56.5%	25.4%	73.0%	29.3%	49.8%	13.8%	69.6%
Mistral 7B	7B	62.5%	81.0%	74.2%	82.2%	80.5%	54.9%	23.2%	62.5%	26.2%	50.2%	12.7%	50.0%
Mixtral 8x7B	13B	70.6%	84.4%	77.2%	83.6%	83.1%	59.7%	30.6%	71.5%	40.2%	60.7%	28.4%	74.4%

Assignment 4

- Go to <https://people.eecs.berkeley.edu/~hendrycks/data.tar>
 - Use the Val dataset that has ground truth
- Colab should have both mixtral and mistral 7B pipelines
- Compare both in terms of speed, accuracy
- Bonus: 100 pts
 - Apply SMoE to other models
- Due 9/3

Team Project Presentations

Be Amazing!!

